

# FIT9132 - Database Assignment 1

**Student ID:** 35501308

**Student Name:** Bharath Arun Gandhimani

## TASK 2 - Ocean Odyssey Normalization

**Cruise ID** 1234  
**Cruise Name** Sydney Brisbane Getaway  
**Cost per person** \$810

| Port Code | Port Name | Port Latitude | Port Longitude | Country code | Country Name | Departure/Arrival | Date/Time         |
|-----------|-----------|---------------|----------------|--------------|--------------|-------------------|-------------------|
| AUSYD     | Sydney    | -33.865143    | 151.209900     | AUS          | Australia    | Depart            | 19 Feb 2025 15:30 |
| AUBNE     | Brisbane  | -27.470125    | 153.021072     | AUS          | Australia    | Arrive            | 21 Feb 2025 07:00 |
| AUBNE     | Brisbane  | -27.470125    | 153.021072     | AUS          | Australia    | Depart            | 21 Feb 2025 17:00 |
| AUSYD     | Sydney    | -33.865143    | 151.209900     | AUS          | Australia    | Arrive            | 23 Feb 2025 08:00 |

UNF:

CRUISE (cruise\_id, cruise\_name, cost\_per\_person, (port\_code, port\_name, port\_latitude, port\_longitude, country\_code, country\_name, departure\_arrival, date\_time))

1NF:

CRUISE (cruise\_id, cruise\_name, cost\_per\_person)

CK:

cruise\_id

ITINERARY (cruise\_id, port\_code, date\_time, port\_name, port\_latitude, port\_longitude, country\_code, country\_name, departure\_arrival)

CK:

cruise\_id, port\_code, date\_time

**Partial Dependencies:**

port\_code -> port\_name, port\_latitude, port\_longitude, country\_code, country\_name

2NF:

CRUISE (cruise\_id, cruise\_name, cost\_per\_person)

ITINERARY (cruise\_id, port\_code, date\_time, departure\_arrival)

PORT (port\_code, port\_name, port\_latitude, port\_longitude, country\_code, country\_name)

**Transitive Dependencies:**

country\_code -> country\_name

3NF

CRUISE (cruise\_id, cruise\_name, cost\_per\_person)

ITINERARY (cruise\_id, port\_code, date\_time, departure\_arrival)

PORT (port\_code, port\_name, port\_latitude, port\_longitude, country\_code)

COUNTRY (country\_code, country\_name)

**Full Dependencies:**

cruise\_id -> cruise\_name, cost\_per\_person

cruise\_id, port\_code, date\_time -> departure\_arrival

port\_code -> port\_name, port\_latitude, port\_longitude, country\_code

country\_code -> country\_name

Final 3NF Relations:

CRUISE (cruise\_id, cruise\_name, cost\_per\_person)

ITINERARY (cruise\_id, port\_code, date\_time, departure\_arrival)

PORT (port\_code, port\_name, port\_latitude, port\_longitude, country\_code)

COUNTRY (country\_code, country\_name)

## Attribute Synthesis:

From Task 1, we have the following relations:

OPERATOR (operator\_id, company\_name, ceo\_name)

SHIP (ship\_code, ship\_name, commission\_date, tonnage, max\_guest\_cap, reg\_country)

CRUISE (cruise\_id, cruise\_name, cruise\_desc, depart\_date, depart\_time)

CABIN (cabin\_no, ship\_code, sleeping\_capacity, cabin\_class)

BOOKING (cruise\_id, pass\_id, cabin\_no, boarding\_time, boarding\_date)

PASSENGER (pass\_id, guardian\_id, pass\_fname, pass\_lname, pass\_gender, pass\_dob, pass\_street, pass\_town, pass\_postcode, pass\_country)

From Task 2, we have the following relations:

CRUISE (cruise\_id, cruise\_name, cost\_per\_person)

ITINERARY (cruise\_id, port\_code, date\_time, departure\_arrival)

PORT (port\_code, port\_name, port\_latitude, port\_longitude, country\_code)

COUNTRY (country\_code, country\_name)

We can notice that the relation “CRUISE” happens to be in both Task 1 and Task 2 relations. So we need to merge these two identical relations.

Therefore,

## FINAL 3NF Relations:

OPERATOR (operator\_id, company\_name, ceo\_name)

SHIP (ship\_code, ship\_name, commission\_date, tonnage, max\_guest\_cap, reg\_country)

CRUISE (cruise\_id, cruise\_name, cruise\_desc, depart\_date, depart\_time, cost\_per\_person)

CABIN (cabin\_no, ship\_code, sleeping\_capacity, cabin\_class)

BOOKING (cruise\_id, pass\_id, cabin\_no, boarding\_time, boarding\_date)

PASSENGER (pass\_id, guardian\_id, pass\_fname, pass\_lname, pass\_gender, pass\_dob, pass\_street, pass\_town, pass\_postcode, pass\_country)

ITINERARY (cruise\_id, port\_code, date\_time, departure\_arrival)

PORT (port\_code, port\_name, port\_latitude, port\_longitude, country\_code)

COUNTRY (country\_code, country\_name)